
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) models are now a central mechanism for scaling large language models, but the representational mechanisms that distinguish them from dense networks remain under-specified. Prior work characterizes how dense models use *superposition* to encode more features than available dimensions, and how the resulting interference depends on feature sparsity and feature importance. We show that this explanatory frame does not transfer directly to MoEs. In our toy-model setting, neither feature sparsity nor feature importance induces discontinuous phase changes in MoE experts; instead, network sparsity, defined by the ratio of active to total experts, provides the relevant control variable. We introduce metrics for quantifying superposition across experts and find that higher network sparsity is associated with greater *monosemanticity*. We further define expert specialization in terms of monosemantic feature representation rather than aggregate load balancing, and show that appropriately initialized experts organize around coherent feature combinations. These results indicate that network sparsity can improve the interpretability of MoE representations while preserving comparable reconstruction performance, thereby refining the assumed tradeoff between interpretability and capability.

1 INTRODUCTION

Mixture of Experts (MoEs) are widely used in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), because conditional computation yields favorable efficiency–performance tradeoffs (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work refines the routing procedure (e.g., Expert-Choice routing) and improves training stability and transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Complementary theoretical and empirical analyses show that learned routers can recover latent cluster structure in the data distribution, offering one explanation for expert specialization (Dikkala et al., 2023). Nevertheless, the internal representational mechanisms of MoEs remain substantially less characterized from a mechanistic-interpretability perspective.

Interpretability-oriented MoE methods attempt to make expert behavior more legible by modifying the architecture or routing objective. Yang et al. (2025b) propose MoE-X, which promotes sparsity-aware routing and uses wide ReLU experts to reduce polysemanticity. Park et al. (2025) introduce MONET, which scales the number of experts so that capability editing can be mediated by expert activation. These methods primarily intervene at the architectural level, leaving open the mechanistic questions of how MoEs encode features, how expert partitioning changes superposition, and whether specialization emerges without additional regularization. Mu & Lin (2025) accordingly identify mechanistic interpretability as a central unresolved problem in MoE research.

A fundamental obstacle for neural-network interpretation is superposition, the regime in which a model represents more features than it has available dimensions. This mechanism permits sparse features to be packed into a smaller set of neurons, but it also induces polysemantic neurons whose activations no longer correspond cleanly to individual features.

MoE architectures add an additional control variable to this representational problem: network sparsity. In contrast to dense models, which apply all parameters to every input, MoEs activate only a subset of the available experts for each example (Shazeer et al., 2017). Dense models therefore exploit feature sparsity by compressing many rarely active features into shared dimensions, whereas MoEs can use routing to allocate expert capacity selectively to particular feature combinations.

We study three questions: (1) whether MoEs exhibit less superposition than parameter-matched dense models; (2) whether MoE experts display discrete phase changes in the superposition of a target feature as its relative importance and the global feature sparsity vary, as observed in dense toy models; and (3) whether expert specialization can be formalized in terms of feature representation rather than only aggregate load balancing.

We address these questions using controlled toy models that extend the framework of Elhage et al. (2022) to MoE architectures. Our contributions are: (1) MoEs do not exhibit the sharp phase changes observed in dense models, instead showing increasingly continuous transitions as network sparsity increases; (2) MoEs consistently exhibit greater monosemanticity, equivalently lower superposition, than dense models with matched active and total parameter budgets; and (3) we introduce an interpretability-oriented definition of expert specialization based on monosemantic feature representation, under which experts organize around coherent feature combinations rather than arbitrary load-balancing partitions.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis posits that neural networks encode concepts as directions in activation space (Park et al., 2024), while the number of interpretable features may exceed the dimensionality of that space. Elhage et al. (2022) formalized this overcomplete regime as superposition and showed that dense models use non-orthogonal feature packing to increase representational capacity at the cost of polysemanticity. Although superposition can be capacity-efficient in theory (Scherlis et al., 2025), it often requires post-hoc disentanglement methods, including Sparse Autoencoders, to recover monosemantic features (Bricken et al., 2023). Recent studies analyze how data correlations modulate superposition (Prieto et al., 2025) and how interference patterns arise or can be reduced (Gurnee et al., 2023). We extend this line of analysis to MoE architectures and show that network sparsity, rather than feature sparsity alone, is a primary determinant of the learned representational strategy.

Interpretability of MoEs. Although MoEs are a standard scaling technique for large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), their internal representations are less mechanistically understood than those of dense architectures. Existing analyses primarily characterize macroscopic quantities, including routing stability (Zoph et al., 2022), expert-selection statistics (Zhou et al., 2022), and latent cluster recovery (Dikkala et al., 2023). Recent interpretability-focused methods impose specialization through architectural constraints, such as predefined concept routing (Yang et al., 2025b), or by increasing the expert count to support vocabulary-scale capability editing (Park et al., 2025). These approaches clarify useful interventions but do not directly characterize intrinsic feature geometry. We address this gap by measuring how expert routing changes superposition, showing that MoEs are more monosemantic than dense baselines, and formalizing specialization as a relation between routed input directions and monosemantic feature representations.

3 BACKGROUND

A central objective of mechanistic interpretability is to reverse engineer neural networks by decomposing internal representations into human-interpretable concepts, or ‘features’. A common assumption is that these features are linearly represented, so that a hidden state h can be expressed as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

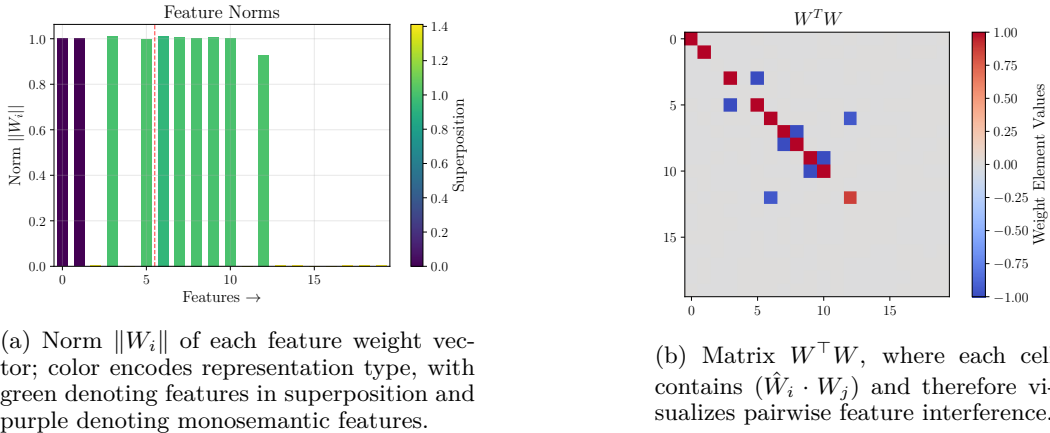
where $\vec{f}_i$ denotes the representation direction for feature i , α_i denotes the corresponding activation coefficient, approximately measuring the presence of feature i in the input, and F is the set of represented features.

The *superposition hypothesis* states that a model can represent substantially more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Such overcomplete representation requires feature vectors in F to be packed non-orthogonally. When two feature directions satisfy $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, their representations interfere, and we describe the features as being in *superposition*. This configuration can be effective when features are sparsely active, as is typical for text data, but it reduces interpretability because α_i can include spurious contributions not caused by the presence of feature i .

Monosemanticity is typically defined at the neuron level: a monosemantic neuron has an activation that corresponds cleanly to a single coefficient α_i , equivalently to a basis-aligned feature. Orthogonal but non-basis-aligned features can have low interference while still producing *polysemantic* neurons. In this paper, our primary object of measurement is the reduction of superposition rather than strict basis alignment. Accordingly, when we describe models, experts, or features as “more monosemantic,” we mean that their feature representations exhibit lower superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are often described as ensembles of dense subnetworks, with each expert treated as an independent dense model evaluated conditionally on the input. This description does not specify whether individual experts learn feature representations with the same interference structure as dense networks. We therefore compare dense and MoE models directly through their measured superposition.



(a) Norm $\|W_i\|$ of each feature weight vector; color encodes representation type, with green denoting features in superposition and purple denoting monosemantic features.

(b) Matrix $W^T W$, where each cell contains $(\hat{W}_i \cdot W_j)$ and therefore visualizes pairwise feature interference.

Figure 1: Feature representation and superposition for a dense model with $n = 20$ features, $m = 6$ hidden dimensions, importance $I = 0.7^i$, and uniform feature density $(1 - S) = 0.1$. Color represents the superposition score $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

We study an MoE autoencoding problem in which a high-dimensional input vector $x \in \mathbb{R}^n$ is projected into lower-dimensional expert states $h \in \mathbb{R}^m$ and then reconstructed. This construction extends the toy-model framework of Elhage et al. (2022) from dense autoencoders to conditionally routed experts.

The input distribution. The input vector x represents activations in an idealized disentangled representation: each coordinate x_i is a distinct independent feature, corresponding to a perfectly neuron-aligned and monosemantic basis. For nonzero activations, we sample

$x_i \sim U(0, 1)$, and we impose feature sparsity $S \in [0, 1]$ by setting $P(x_i = 0) = S$. Because features need not contribute equally to the objective, each feature x_i receives a scalar importance weight I_i in the reconstruction loss. To isolate phase-transition behavior for a single target feature, we scale the last feature by $r \in \mathbb{R}^+$ and set $I = (1, 1, \dots, r)$. The variables I_i and r therefore parameterize feature signal strength and allow us to measure expert sensitivity to feature magnitude.

Model Architecture. The MoE contains E experts. Each expert e is parameterized by a projection matrix $W^e \in \mathbb{R}^{m \times n}$ and an output bias $b^e \in \mathbb{R}^n$. A learned router computes $g(x) = \text{softmax}(W^r x)$ with $W^r \in \mathbb{R}^{E \times n}$, and the input is assigned to the top- k experts under this distribution. Each selected expert maps the input to a hidden state $h^e = W^e x$ and produces the reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The model output is the gating-weighted sum of active expert reconstructions, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where w_e denotes the renormalized top- k gating weight.

Training minimizes an L2 reconstruction objective weighted by feature importance, I_i , namely $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, we additionally include the standard auxiliary load-balancing term (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$, to reduce expert collapse. Here, N is the number of experts, f_e is the fraction of batch elements routed to expert e , P_e is the average router probability assigned to expert e across the batch, and $\alpha = 0.01$ sets the penalty strength. We omit this auxiliary term in Section 4 so that the measured superposition reflects the intrinsic architectural bias of the MoE rather than an explicit balancing regularizer.

4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we measure two quantities: *representation strength* and *interference* with other features. Representation strength for feature i in expert e is the weight-vector norm $\|W_i^e\|$. Values near one indicate that feature i is fully represented in expert e , whereas values near zero indicate that the feature is not learned by that expert. We visualize feature interference within expert e using the Gram matrix $W^\top W$, whose off-diagonal entries encode pairwise feature interactions.

Figures 1a and 2a show that, under equal total parameter count ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$), the dense model and MoE represent comparable numbers of features (10 versus 8) with similar norm magnitudes. The MoE experts still exhibit local superposition, for example in Expert 0 of Figure 2b, but the global interference pattern is partitioned by routing. In the dense model, any represented feature can interfere with any other represented feature (Figure 1b); in the MoE, features assigned to different experts have zero cross-expert interference, producing a block-diagonal structure. This result indicates that MoEs allocate representational capacity by partitioning feature space and thereby restricting the scope over which interference can occur.

Expert Feature Dimensionality. To characterize how MoEs allocate limited representational capacity relative to dense models, we measure feature dimensionality, defined as the “fraction of a dimension” assigned to a feature (Elhage et al., 2022). For feature i , its dimensionality in expert e is

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e ranges from zero, corresponding to an unlearned feature, to one, corresponding to a monosemantic representation. The aggregate MoE feature capacity is therefore $D = \sum_e \sum_{i=1}^n D_i^e$.

¹The output ReLU enforces non-negative reconstructions, matching the input support $x_i \in [0, 1]$ for nonzero features. In addition, off-diagonal terms in $(W^e)^\top W^e$ can induce negative interference, and ReLU suppresses those negative components. When features are sparse, negative interference is effectively filtered to zero, favoring parameter configurations with negative off-diagonal terms (e.g., antipodal pairs).

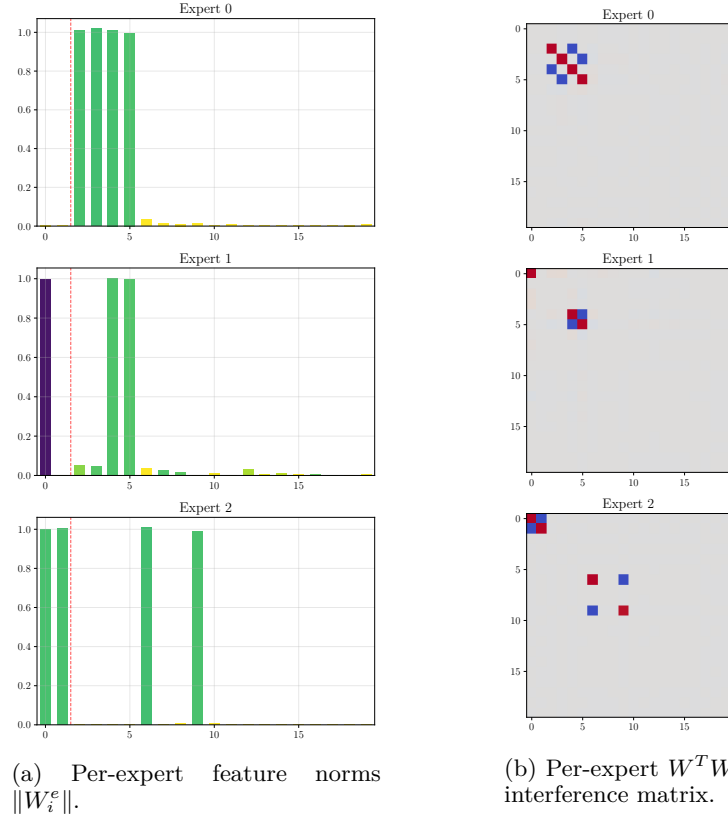


Figure 2: Feature representation and superposition for a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under top- $k = 1$ routing, with importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

Efficient Packing. Efficient feature packing implies that total feature dimensionality approximately equals the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE, the corresponding capacity relation is $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, indicating that MoEs pack features with comparable efficiency to dense models when total parameter count is matched.

Features per Dimension & Network Sparsity. Because both dense and MoE models efficiently pack features, we compare architectures by the number of represented features per dimension. This statistic directly measures superposition and permits evaluation of how expert count changes superposition across feature-sparsity regimes. Values greater than one indicate superposition, since the model represents more features than dimensions. For a MoE, features per dimension is defined as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ denotes the Frobenius norm and p_e is the expected usage probability of expert e over a batch, equivalently the average renormalized gating weight after top- k routing.

For MoEs with the same total parameter count as the dense model, the dense model has more features per dimension (Figure 3) and therefore more superposition. This implies that the dense model uses superposition to represent a larger number of learned features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE tends to remain near the monosemantic capacity limit ($n_{\text{learned}} \approx m_{\text{total}}$). As the total number of experts increases while holding total parameters

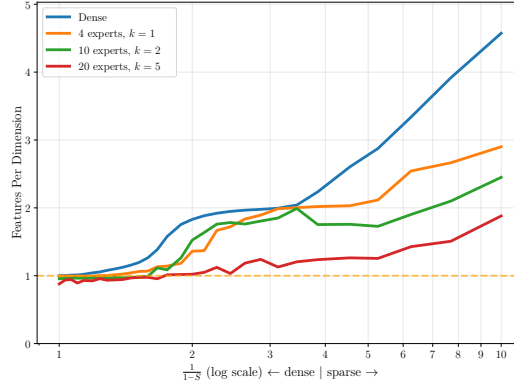


Figure 3: Features per dimension as a function of inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the highest superposition. Superposition decreases as expert count increases: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameter counts and similar k/E . The dashed line at 1.0 denotes monosemantic representation.

and the ratio k/E approximately fixed, features per dimension decreases. Equivalently, increasing the number of experts reduces superposition and yields more monosemantic feature representations. The dense model’s greater superposition also produces consistently lower reconstruction loss than the MoEs in Figure 6 of Appendix A.1, with a difference of $\sim 0.03 - 0.08$ at a fixed sparsity level. However, as expert count increases, MoE reconstruction loss approaches the dense-model loss. Appendix A.2 provides theoretical intuition for this pattern.

5 PHASE CHANGE

Although MoEs and dense models learn comparable numbers of features, MoEs allocate those features across experts with reduced interference. This indicates that network sparsity primarily changes the allocation geometry of features rather than only the number of features learned. We therefore evaluate whether input-distribution parameters, including feature sparsity and feature importance, determine this allocation and whether they induce physics-inspired *phase changes* in the learned representation.

In these toy models, each feature can occupy one of a finite set of representational regimes: it may be ignored, represented in superposition, or represented monosemantically. A phase change denotes a discontinuous boundary between such regimes as a function of feature sparsity and relative feature importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature-representation strategies (Elhage et al., 2022). Varying feature sparsity and relative importance in the input distribution changes the learned solution; for example, increasing sparsity can make superposition more favorable. We construct expert-specific phase diagrams for MoEs to test whether routed experts follow the same regime boundaries as dense models or instead adopt distinct allocation strategies.

Setup. We use the experimental setup from Section 4.1, adding the load-balancing loss in configurations where it is needed to prevent specialization collapse, i.e. local optima in which the router ignores one or more experts. We evaluate three architectures, each with one active expert ($k = 1$): (A) $n=2$, $m=1$; and (B) $n=3$, $m=1$; and (C) $n=3$, $m=2$. For each architecture, we report expert-specific phase diagrams over feature sparsity and last-feature relative importance while varying network sparsity by increasing the number of experts (E) up to the input dimensionality (n).

In this section, we hold the number of active parameters fixed rather than the total parameter count, so that $m_{dense} = km_{moe}$ up to router parameters. This design isolates comparisons within an architecture; otherwise, observed differences could be attributed to changes in active capacity rather than routing structure. Incidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. However, the latter uses a single hidden dimension ($m = 1$) to encode the same number of input features ($n = 3$) that the former encodes with two hidden dimensions ($m = 2$), which complicates a direct superposition-based interpretation of specialization.

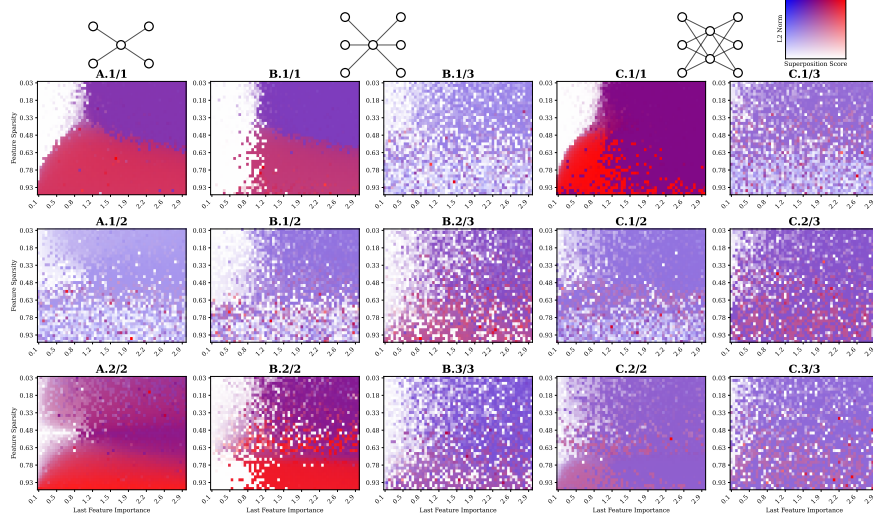


Figure 4: Expert-specific embedding regime for the final input dimension. For a fixed expert and feature, the visualization classifies the feature as ignored (white), monosemantic (blue-purple), or represented in superposition (red). Each cell combines feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) over feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where n denotes the final feature among n total features. For each cell, ten models are trained and the minimum-loss run is selected. Load-balancing loss is used throughout this section. Low L2 norm ($\|W_n\|$) indicates that the feature is ignored; otherwise, low superposition is shown in blue-purple and high superposition in red. Cell (i, j) in subfigure X.e/E corresponds to expert e of E total experts for architecture X, last-feature importance $I_n = i$, and sparsity $S = j$; X.1/1 denotes the dense model.

Results and Takeaways. All single-expert, dense cases exhibit clear phase changes (Figure 4.X.1/1), consistent with Elhage et al. (2022). As the number of experts increases, these discrete boundaries disappear. Some $E = 2$ MoE experts resemble the corresponding dense cases (Figure 4.X.2/2), but their transitions are more continuous. In each architecture, one expert becomes more monosemantic and specializes in the feature with highest relative importance. Experts that do not resemble the dense baseline have uniformly lower superposition scores, shown by bluer cells, indicating more monosemantic representations. This pattern is consistent with the preceding section: relative to dense models, MoEs favor lower-superposition solutions.

For the $n = 2, m = 1$ architecture (Figure 4.A), the dense model ignores the final feature at low feature sparsity. The comparable MoE preserves that feature more strongly because routed capacity can be allocated to it. With three input dimensions (Figure 4.B), this preservation effect is absent because experts use the single hidden dimension to superimpose the other two features, leaving no remaining monosemantic capacity for the third feature. In architecture B, unlike the other two cases, one hidden dimension with superposition is insufficient to represent all features in the high-sparsity regime. Even so, we do not observe a sharp phase change, except in the feature-sparsity interval 0.5–0.7 in 4.B.2/2, where the transition is largely discrete but mixed. For $m = 2$, the dense model’s white region in Figure 4.C.1/1, corresponding to mid-to-low feature sparsity and lower relative importance of the

final feature, indicates that the final feature is ignored. As network sparsity increases across the remaining Figure 4.C panels, the final feature is represented with larger L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Thus, dimensionality in the low-relative-importance regime increases with network sparsity, matching the trend in Figure 3.

Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 contain an intermediate sparsity window, approximately 0.48 to 0.7, in which polysemantic representation, monosemantic representation, and feature omission coexist. This mixed regime suggests that MoEs can attain comparable loss with either polysemantic or monosemantic solutions, making optimization less likely to select a single consistent strategy in the intermediate regime. Dense models do not show the same pattern, providing evidence that routed experts learn representational strategies distinct from dense networks.

Conclusion. All phase-change experiments use top- $k = 1$, so each input activates a single expert. Despite this constraint, MoEs behave substantially differently from dense models even at $E = 2$, including cases where hidden-dimensional capacity with superposition is sufficient to represent all features. These results indicate that MoEs should not be treated as simple aggregates of dense models. The router restricts each expert to a subset of the feature domain, and this conditional observation mechanism substantially changes both expert optimization and the resulting representations.

6 EXPERT SPECIALIZATION

Because MoEs exhibit lower superposition, we next analyze how monosemantic features are organized across experts and how this organization should be interpreted as specialization.

Existing definitions of MoE specialization often emphasize load balancing across experts over the full input distribution (Chaudhari et al., 2025). This aggregate criterion does not capture the more local notion that an expert should be activated specifically when the concepts it represents are present in the input.

We define an expert as specialized when it both *occupies* particular feature directions in input space and represents those same features relatively monosemantically. We then show that these two conditions are empirically correlated and that their conjunction can induce balanced expert usage as a consequence of feature-conditional routing.

Because we fix $k = 1$, the router partitions feature space into convex cone regions (Appendix A.3), each assigned to a single expert. By definition, cone membership satisfies $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points in the cone and s is any positive scalar. Thus, if a feature vector $\vec{x}$ is routed to expert e , every positive rescaling $s\vec{x}$ is routed to the same expert. We say in this case that feature x is contained in expert e , or equivalently that expert e *occupies* x .

Empirically, small models that distribute input space across more experts tend to obtain lower reconstruction loss (Appendix A.4). This observation motivates the following question: when the router allocates an input-space region to an expert, does that allocation predict the expert’s internal feature representation? Under our definition, specialization predicts that occupied directions should be represented more monosemantically, a correlation that we test in larger toy models (e.g., $m = 10$).

For models with $n > 2$, we test whether initializing experts to occupy specified input-space feature directions makes those experts more monosemantic with respect to the same features. Conversely, for features that an expert learns to represent monosemantically, we test whether the router assigns the corresponding input directions to that expert.

With diagonal gate initialization, each expert monosemantically represents exactly the single feature it initially occupies (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of its five initially occupied features (Figure 5b). Experts initialized on later feature blocks do not monosemantically represent those less important features, nor do they represent the five highest-importance features that they were not initialized to occupy. When feature-importance order is randomized across the initially

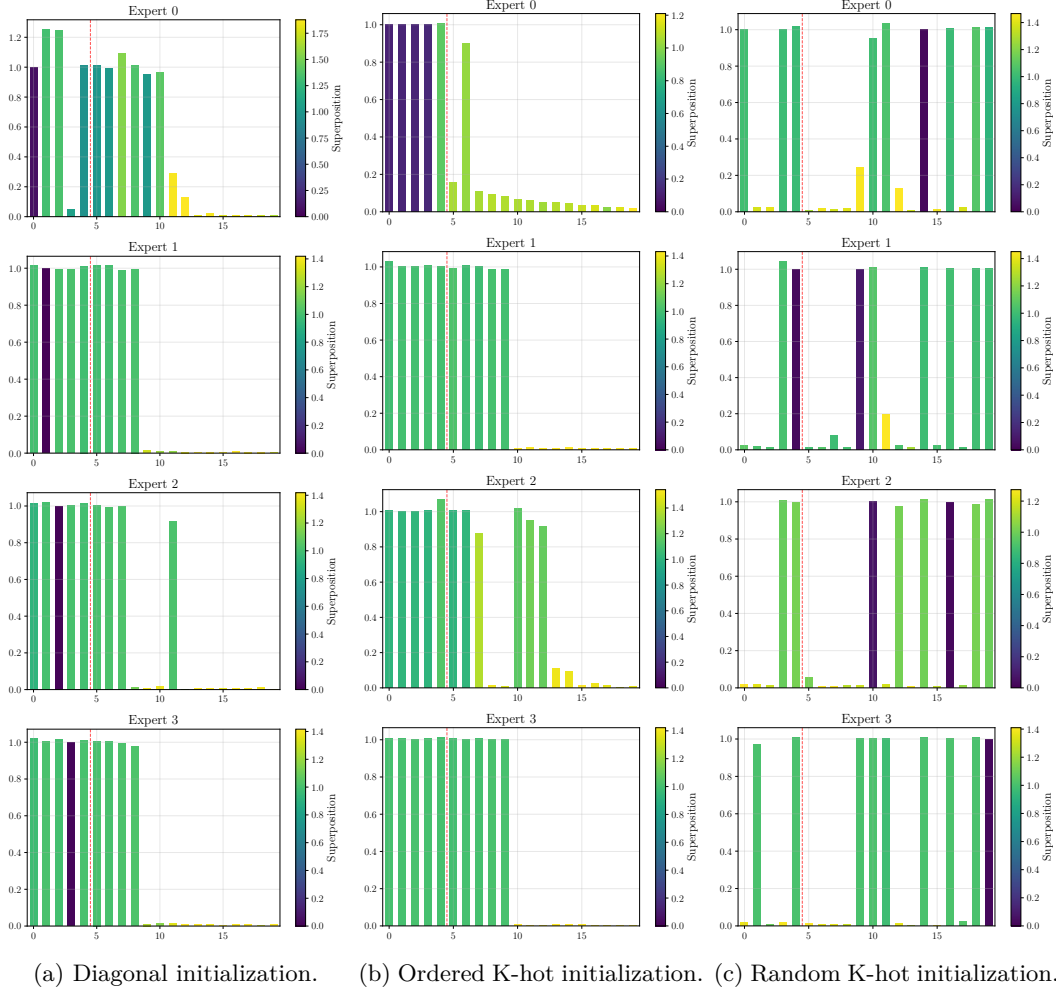


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition scores, encoded by color, under three initialization schemes with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized on the main diagonal ($W_i^T = \hat{e}_i$, the basis vector for that dimension), and relative feature importance decreases exponentially from feature one through feature 20. In (b), the gate matrix uses an “ordered k-hot” initialization: the first expert is aligned with the first five features, and each subsequent expert is aligned with the next block of five features. Relative feature importance matches (a). In (c), the gate matrix uses a “random k-hot” initialization, assigning five random features to each expert such that experts share no feature but collectively cover all 20 features. Relative feature importance decreases exponentially but is randomly permuted across features.

occupied feature sets, each expert monosemantically represents only the most important feature or features within its assigned set (Figure 5c).

These results show a strong correspondence between the features initially routed to an expert and the features that the expert ultimately represents monosemantically. They also show that monosemantic representation is selective for high-importance features. This selectivity appears both when each expert is explicitly initialized with one important feature and when each expert receives a set of candidate features from which it effectively selects the most important feature to represent monosemantically.

Under uniform feature importance, experts place all features into higher-interference superposition. Even in this setting, the features initially occupied by an expert remain *relatively*

more monosemantic: their superposition scores are, on average, 1.03 standard deviations below that expert’s mean superposition score.

We next measure whether monosemantic representation of a feature by an expert correlates with router occupation of the corresponding input direction. We compute expert-usage statistics in two conditions: when the expert’s monosemantic feature or features are *one of many* active features, and when they are the *only* active features. The latter condition directly estimates the probability that the expert occupies those feature directions. Table 1 shows that this correlation holds under both xavier and k-hot initialization. Because $E = 10$, mean usage of $\sim 10\%$ corresponds to uniform load balancing. In every case, expert usage increases substantially when the expert’s monosemantic feature set is active; when that feature set is the only active input content, the corresponding expert dominates routing. Under k-hot initialization, 100% of features represented monosemantically by an expert are also occupied by that same expert.

Table 1: Per-expert monosemantic feature counts and usage statistics for $n = 100, m = 10, E = 10$. For each initialization scheme (xavier and k-hot), one hundred models are trained, yielding 1000 total experts per scheme. Statistics are aggregated across models after classifying experts by the number of features they represent monosemantically. For each expert’s monosemantic feature set, we report expert usage when those features are one of several active input features and when those features are the *only* active input features.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

Experts that represent more features monosemantically satisfy a stronger specialization criterion. Their marginal usage on arbitrary inputs decreases, but their conditional usage increases substantially when their specialized feature set is active. This pattern holds in all cases except the xavier-initialized model containing an expert with four monosemantic features, where utilization drops markedly.

7 CONCLUSION

We analyzed how expert routing changes superposition in MoEs and found that, in controlled toy autoencoders, MoEs consistently produce more monosemantic representations than dense networks while eliminating the sharp phase changes observed in dense baselines. We also introduced a feature-based definition of expert specialization and showed that, under suitable initialization, experts organize around coherent feature directions. Because these results are obtained in synthetic autoencoder settings, their transfer to large-scale

transformers with unknown feature distributions remains an open question (Appendix A.6). Within this setting, however, toy MoEs obtain comparable reconstruction loss while preserving more interpretable representations, indicating that interpretability and capability need not be intrinsically opposed. Future work should identify the conditions that favor monosemanticity in MoEs, characterize how MoE training dynamics differ from dense-model training dynamics, and specify when feature-level specialization emerges.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

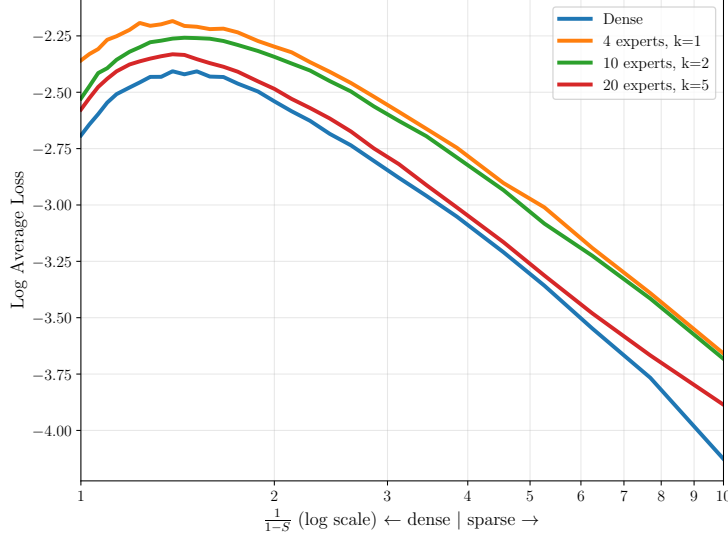


Figure 6: Log average loss as a function of feature density ($\frac{1}{1-s}$) for dense ($m = 20$), MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models with uniform feature importance ($I_i = 1.0$) and $n = 100$ input features. Results are averaged over five runs per sparsity level. The dense model has lower loss than all MoE variants at every sparsity level, but MoE loss approaches dense-model loss as the number of experts increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition exploits the mismatch between sparse feature activation and dense computation by compressing rarely active features into shared dimensions. MoEs address the same mismatch structurally through conditional computation. By aligning parameter activation with feature sparsity, MoEs reduce the cost of maintaining dedicated, rarely active feature representations. As noted by Elhage et al. (2022), when computation is expended only on active features, replacing polysemantic neurons with dedicated monosemantic units can become optimal; in this sense, routing substitutes *selection* for the *compression* provided by superposition.

Geometrically, this architectural change reduces *interference*. Even when the global ratio of represented features to parameters is held fixed, the router partitions feature space so that a feature routed to expert e competes only with the subset of features assigned to the same expert. Interference is therefore determined by the local expert matrix $(W^e)^\top W^e$ rather than by the dense model’s global matrix $W^\top W$. This partitioning reduces the number of possible interfering features for each feature vector, decreases the optimization pressure to pack features in superposition, and permits experts to learn more monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

When $k = 1$, the router $g(x) = \text{softmax}(W^r x)$ induces the same expert choice as $\text{argmax}(W^r x)$. The region routed to expert i is characterized by $\forall j \neq i, (w_i - w_j)^\top x > 0$, where w_i and w_j are rows of W^r . Each constraint is a homogeneous linear inequality, and intersections of such inequalities define convex cones. If x satisfies the inequalities, then any positive rescaling sx also satisfies them. Similarly, if x_1 and x_2 satisfy the inequalities, then every convex combination $x = x_1 \lambda + (1 - \lambda)x_2$ with $\lambda \in [0, 1]$ satisfies them as well.

For $k > 1$, the input region associated with a given expert e becomes a union of convex cones. Because a union of convex cones is not generally a convex cone, the geometric interpretation of expert occupation of feature directions need not extend beyond the top-1 routing regime.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

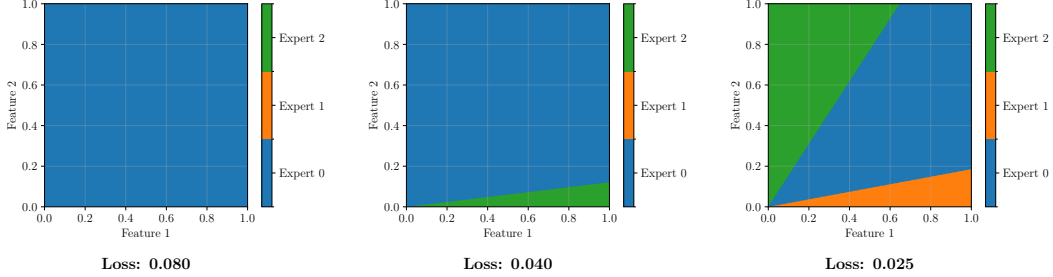


Figure 7: Expert routing for three identical architectures under different initialization schemes, with $n = 2$, $E = 3$, and $m = 1$. The first model (left) has the highest loss (0.08) and routes all inputs to a single expert. The second model (middle) has lower loss (0.04) and routes a small input subset, specifically cases in which feature 1 is active, to a second expert. The third model (right) has the lowest loss (0.025) and partitions the input space across all experts: one expert is selected when only feature 1 is active, one when only feature 2 is active, and one when both features are active.

In small models ($n = 2, m = 1, E > 1$), evaluated over $E \in [2, 7]$, lower loss is associated with routing schemes that distribute input space across more experts. Fixing $n = 2$ permits direct visualization of the input-space regions assigned to each expert, as shown in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setting of single-layer, single-nonlinearity, top- $k = 1$ MoEs, a theoretical correspondence exists between any dense model and a monosemantic MoE with an equivalent number of active features, provided a sparsity constraint holds.

Assume that every input has at most a active features, so that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Also assume that this upper bound is no larger than an expert’s hidden dimensionality m , and that m is no larger than the number of input features n , giving $a \leq m \leq n$. To construct the monosemantic MoE, introduce one expert for each subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$; that expert monosemantically preserves the features in S . It is sufficient to instantiate only subsets with $|S| = a$. The router then selects the expert associated with the active feature set, which contains at most a elements by assumption:

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Because $|S| \leq a \leq m$, each expert has enough hidden dimensionality to represent its assigned feature subset without superposition. Since at most a features are active and each active-feature combination is assigned a dedicated expert, all valid inputs can be represented monosemantically under this construction. The number of possible experts required is therefore $\binom{n}{a}$.

This theoretical MoE has zero reconstruction loss only when the sparsity constraint holds, or in these toy models as sparsity approaches one. If more than m features can be active simultaneously ($a \not\leq m$), the active set exceeds the monosemantic capacity of a single expert. A dense polysemantic model faces the same capacity issue unless the data distribution contains exploitable feature correlations. Thus, even when $a \not\leq m$ occurs occasionally, the monosemantic MoE constructed above may still outperform a polysemantic dense model under relaxed sparsity constraints.

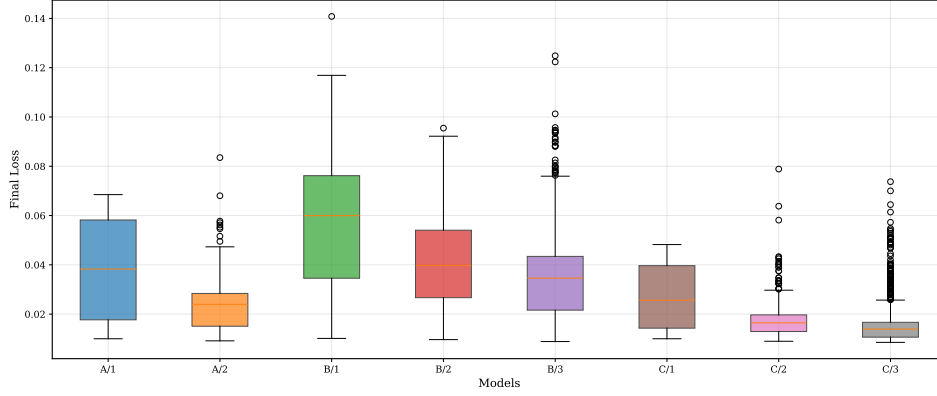


Figure 8: In model notation X/E , X denotes the architecture used in Figure 4 and E denotes the total number of experts, i.e. network sparsity. Increasing network sparsity reduces mean loss while increasing localized variance, especially when the number of experts approaches the number of input feature dimensions. This variance is attributable to the comparatively unstable training dynamics of MoEs relative to dense models, despite training ten models per cell and selecting the minimum-loss run.

Therefore, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ satisfying the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for every valid input. In the toy settings considered here, this sparsity constraint is approached in the limit as sparsity goes to one. In practice, the semantic content of a data source may impose an effective upper bound on the number of meaningful features that must be encoded, as reflected by the dimensionality of useful embeddings. A MoE with sufficiently many experts and a tractable amount of residual superposition may therefore be able to encode the relevant feature set while remaining interpretable.

A.6 LIMITATIONS

Several limitations constrain the interpretation of these results. First, all experiments are conducted in controlled toy-model settings derived from sparse-autoencoder studies of superposition (Elhage et al., 2022). This design enables precise measurement of interference and monosemanticity, but it removes many properties of large-scale transformers, including multi-layer computation, attention, heterogeneous feature distributions, and realistic routing dynamics. Consequently, it remains uncertain how directly these patterns transfer to deployed MoE architectures.

Second, the architectural and routing choices are deliberately restricted: the experiments use simple experts, fixed top- k routing (often $k = 1$), matched parameter budgets between dense and MoE models, and no auxiliary load-balancing loss except where explicitly stated. Practical MoEs commonly use richer routing mechanisms, variable expert capacities, and multi-task or language-modeling objectives (Lepikhin et al., 2021; Fedus et al., 2022); moving from reconstruction to next-token prediction, for example, may induce qualitatively different representational behavior.

Third, the synthetic feature distributions used here, including their sparsity patterns, importance weights, and independence assumptions, are substantially simpler than the feature distributions expected in natural data.

Finally, although MoEs in our experiments exhibit increased monosemanticity and reduced superposition under fixed conditions, we do not evaluate downstream performance tradeoffs, large-scale training stability, or specialization dynamics in full transformer models. Prior work suggests that these dynamics can vary substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

810 Overall, the study provides mechanistic evidence under clean experimental conditions, but
811 additional work is required to validate the same representational patterns in large, realistic
812 MoE systems. The present measurement methods do not scale directly to such models,
813 making scalable diagnostics for superposition and specialization an important direction for
814 future work.